

```
PS C:\Users\moham\Documents\MongoDB_Project> mongosh ecommerce_mongodb.js
```

```
All collections dropped. Starting fresh build...
```

```
All collections created with schema validators.
```

```
All indexes created.
```

```
— Inserting users...
```

```
Inserted 20 users.
```

```
— Inserting categories...
```

```
Inserted 15 categories.
```

```
— Inserting sellers...
```

```
Inserted 15 sellers.
```

```
— Inserting products...
```

```
Inserted 20 products.
```

```
— Inserting coupons...
```

```
Inserted 15 coupons.
```

```
— Inserting orders...
```

```
Inserted 18 orders.
```

```
— Inserting payments...
```

```
Inserted 18 payment records.
```

```
— Inserting reviews...
```

```
Inserted 20 reviews.
```

```
— Inserting carts...
```

```
Inserted 15 carts.
```

```
— Inserting inventory records...
```

```
Inserted 20 inventory records.
```

ALL DATA INSERTION COMPLETE

SECTION 4 – CRUD OPERATIONS

— 4.1 INSERT OPERATIONS —

insertOne: Adding new user 'Tom Hanson'...
insertOne (users) complete.

insertOne: Adding flash coupon 'CYBER25'...
insertOne (coupons) complete.

insertMany: Adding 3 new products...
insertMany (products) complete.

— 4.2 READ OPERATIONS —

find: All active customers:
find (users) complete.

find + projection: Product names, prices, and ratings:
find + projection complete.

findOne: Retrieve order ORD-2024-000001:
findOne (orders) complete.

find: All delivered orders:
find (delivered orders) complete.

— 4.3 UPDATE OPERATIONS —

updateOne: Mark order ORD-2024-000013 as 'shipped' and add tracking number...
updateOne (orders) complete.

updateOne: Award 250 loyalty points to Alice Johnson...
updateOne (users - loyalty points) complete.

updateOne: Deactivate expired coupon FLASH40...
updateOne (coupons) complete.

updateOne: Add new storage variant to Samsung T9 SSD...
updateOne (\$push variant) complete.

updateMany: Apply 10% sale price to all book products...
updateMany (products - books sale) complete.

updateMany: Increment usedCount by 1 for all active coupons with usedCount > 0..
updateMany (coupons - usedCount increment) complete.

updateOne: Update delivery address for Bob Martinez...
updateOne (nested address update) complete.

updateMany: Deactivate inventory items with quantity = 0...
updateMany (inventory - zero stock) complete.

— 4.4 DELETE OPERATIONS —

deleteOne: Remove cart for user 'tom.hanson@email.com' (empty cart cleanup)...
deleteOne (empty cart) complete.

deleteMany: Delete fully-used & expired coupons...
deleteMany (expired coupons) complete.

deleteMany: Delete reviews with 0 helpfulVotes that are older than 1 year...
deleteMany (stale reviews) complete.

SECTION 5 – QUERY OPERATIONS

— 5.1 COMPARISON OPERATORS —

\$gt: Products with price > \$500:

\$lt: Products with price < \$50 (budget picks):

\$gte: Users with loyaltyPoints >= 1000:

\$lte: Coupons with maxDiscountAmount <= 50:

\$ne: Orders NOT in 'delivered' status:

\$ne: Inventory items not in warehouse 'WH-NY':

— 5.2 LOGICAL OPERATORS —

\$and: Active electronics products with averageRating >= 4.7:

\$or: Users who are sellers OR admins:

\$and + \$or: Orders (delivered OR shipped) with totalAmount > \$400:

\$or: Products that are featured OR on sale:

— 5.3 \$in AND \$nin —

\$in: Orders with status in ['pending', 'confirmed', 'shipped']:

\$in: Inventory in warehouses ['WH-NY', 'WH-LA', 'WH-CHI']:

\$nin: Products NOT tagged as 'apple' or 'samsung':

\$nin: Coupons that are NOT of type 'percentage':

— 5.4 PROJECTION —————

Projection (include): Product catalog view – name, brand, price, rating:

Projection (exclude): User profile – hide passwordHash and _id:

Projection (nested): User locations – name, city, country only:

Projection: Order summary view – orderNumber, status, totalAmount:

— 5.5 SORTING —————

sort DESC: Products sorted by price (highest first):

sort DESC: Top-selling products by totalSold:

sort DESC: Users by loyalty points (top loyalty members):

sort DESC: Orders by totalAmount (highest value first):

sort multi-key: Reviews sorted by helpfulVotes DESC, then createdAt DESC:

— 5.6 LIMIT AND SKIP (PAGINATION) —————

Pagination Page 1: Top 5 products by averageRating:

Pagination Page 2: Products 6–10 by averageRating:

Pagination: Orders page 1 (5 per page), newest first:

limit(3): 3 most recently created active coupons:

limit(3): Top 3 sellers by rating:

E-COMMERCE DATABASE SCRIPT COMPLETE

Collections: users, categories, sellers, products,
orders, reviews, carts, payments,
coupons, inventory

Section 3: Data Insertion (10 collections)

Section 4: CRUD Operations (insertOne/Many,
updateOne/Many, deleteOne/Many, find)

Section 5: Query Operations (\$gt,\$lt,\$gte,\$lte,\$ne,
\$and,\$or,\$in,\$nin, projection, sort,
limit, skip)